Animal Reproduction Cheatsheet

Asexual Reproduction, Sexual Reproduction & Vertebrate Fertilization

1. Reproduction: Key Definitions

- **Reproduction:** The production of offspring by an asexual or sexual process.
- **Asexual Reproduction:** Genetically identical cells produced by a **single parent** via **mitosis**. Diploid parent → two diploid daughters.
- **Sexual Reproduction:** Haploid gametes from **two parents** fuse to create a single **diploid zygote**.

2. Asexual Reproduction in Animals (4 Types)

A. Fission

- Parent organism splits into two (or more) roughly equal parts.
- Each part develops into a complete, independent organism.
- **Examples:** sea anemones, planaria (flatworms), many single-celled organisms.
- Planaria can be cut in half — each half regenerates the missing portion.

B. Budding

- A small outgrowth (bud) forms on the parent and develops into a new organism.
- The bud is **smaller** than the parent (unlike fission where halves are equal).
- Bud may detach or remain attached, forming a **colony**.
- **Example:** Hydra — bud grows on the side, eventually pinches off.

C. Fragmentation

- A piece breaks off the parent and regenerates into a complete organism.
- Requires significant **regeneration ability**.
- **Example:** Sea stars — a severed arm (with part of the central disc) can regenerate into a whole new sea star.
- Not all animals can do this; many will simply die if fragmented.

D. Parthenogenesis (“virgin birth”)

- An **unfertilized egg** develops into a new organism without sperm.
- Offspring may be haploid or diploid (if the egg's ploidy is restored).
- **Examples:** Komodo dragons, whiptail lizards (all-female species), some sharks, aphids, honeybees (drones are haploid males from unfertilized eggs).
- Honeybee system: fertilized eggs → females (workers/queens); unfertilized eggs → males (drones).
- Whiptail lizards: all female, still perform mating behavior to stimulate egg development.

3. Sexual Reproduction in Animals

Core concept: Two parents each contribute haploid gametes (egg + sperm) that fuse at fertilization to form a diploid zygote. Produces genetic variation through meiosis and recombination.

Hermaphroditism

- **Sequential hermaphrodites:** change sex during their lifetime. Triggered by social/environmental cues.
- Protogyny = female first → male (e.g., clownfish: dominant individual becomes female).
- Protandry = male first → female.
- **Simultaneous hermaphrodites:** possess both male and female reproductive organs at the same time (e.g., earthworms, some fish). Usually still cross-fertilize with a partner.

Fertilization Types

- **External fertilization:** eggs and sperm released into the environment (usually water). E.g., most fish, amphibians. Requires **synchronization** (spawning behavior).
- **Internal fertilization:** sperm deposited inside the female's body. E.g., reptiles, birds, mammals. Higher fertilization success rate but fewer offspring.

Reproductive Strategies: r vs. K

	r-selected	K-selected
Offspring	Many	Few
Parental care	Little/none	High investment
Maturation	Fast	Slow
Examples	Most fish, insects	Mammals, birds
Survival	Low per individual	High per individual

4. Vertebrate Fertilization & Development

Egg Types by Yolk Content

- **Isolecithal:** Little yolk, evenly distributed (e.g., mammals, sea urchins).
- **Mesolecithal:** Moderate yolk concentrated at vegetal pole (e.g., frogs).
- **Telolecithal:** Large amount of yolk at vegetal pole (e.g., birds, reptiles, fish).

Vertebrate Reproductive Categories

- **Oviparity:** Lay eggs externally. Embryo develops outside the mother. E.g., most fish, amphibians, reptiles, all birds, monotremes (platypus, echidna).
- **Ovoviviparity:** Eggs retained inside the mother but embryo nourished by **yolk** (no placenta). E.g., some sharks, some snakes.
- **Viviparity:** Embryo develops inside the mother and is nourished via a **placenta** or similar structure. E.g., most mammals, some sharks.

Mammal Groups

- **Monotremes:** Egg-laying mammals (platypus, echidna). Still nurse young with milk.
- **Marsupials:** Short gestation; young born very undeveloped and continue growing in a **pouch** (e.g., kangaroos, opossums).
- **Placental (Eutherian):** Longest gestation; embryo nourished via placenta until well-developed. Highest parental investment.

Key Reproductive Concepts

- **Estrous cycle:** Most mammals. Females are only receptive during **estrus** ("heat"). Uterine lining reabsorbed if no pregnancy.
- **Menstrual cycle:** Primates (including humans). Uterine lining is **shed** (menstruation) rather than reabsorbed.
- **Delayed implantation:** Fertilized egg is held before implanting in uterus, timing birth to favorable season (e.g., some bats, bears).
- **Induced ovulation:** Ovulation triggered by mating act itself (e.g., cats, rabbits).
- **Spontaneous ovulation:** Ovulation occurs on a regular cycle regardless of mating (e.g., humans).

5. Quick Comparison: Asexual vs. Sexual

	Asexual	Sexual
Parents	One	Two
Genetic variation	None (clones)	High (meiosis + recombination)
Speed	Fast	Slower

Energy cost	Low	High (finding mate, gametes)
Adaptability	Low (vulnerable to change)	High (diverse offspring)
Cell division	Mitosis	Meiosis → Fertilization

Source: Week 9 Biol 123 — Animal Reproduction Lectures